Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

Bab 1

PENDAHULUAN

Bab 2

TINJAUAN PUSTAKA

[11pt]report upennstyle

[nonamebreak,round]natbib

TITLE IN ALL CAPS Author's Name as in Path@Penn Official Graduate Group Name YEAR Supervisor's Typed Name Full Faculty Title and Affiliation Graduate Group Chairperson's Typed Name, Full Faculty Title Committee Member's Typed Name, Full Faculty Title and Affiliation Committee Member's Typed Name, Full Faculty Title and Affiliation Committee Member's Typed Name, Full Faculty Title and Affiliation

Authorâ€™s Full Legal Name

Write your dedication text in italics here.

Write your acknowledgement text here.

Ringkasan

Write your abstract text here.

Daftar Isi

Daftar Tabel

Daftar Gambar

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UNIVERSITY OF PENNSYLVANIA PH.D. DISSERTATION LATEX TEMPLATE

3.1 Introduction

This template aims to fulfill University format requirements when one follows the instructions, resolves errors and warnings, and does not change the style file or font settings. Your graduate group may have additional formatting requirements like specific citation and bibliography styles.[1] Check with your advisor and edit the template as needed. Please consult the Office of the Provost dissertation *Formatting FAQs* for the most up-to-date requirements and policies.

3.2 Getting Started

The template requires TeX Live 2022 and is optimized for pdfLaTeX and the LaTeX editor Overleaf. Other LaTeX editing programs may require you to compile the document more than once. To claim your free Overleaf Pro account, go to the Penn Libraries Overleaf Portal and click on the green "Log in through your institution" button.

Click here if you have a question about Overleaf, and they'll make sure it reaches the right person. Click here for their help page to learn how to use Overleaf and LaTeX. See Chapter **??** for additional resources.

3.2.1 Packages

The template automatically loads the packages it requires to function. See the PackagesLoadedbyTemplate.tex file for a complete list of all the packages loaded by the template. You may manually load other packages if you wish, and Overleaf will give you a warning message when you compile your document if your package conflicts with one of the template's packages. If you receive a warning, you can consult the specified package documentation on CTAN for a possible resolution.

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[1] If you are using another LaTeX template, please make sure it complies with all University formatting requirements.

3.3 Margins

As of January 2023, the margins must be 1 inch on each side. Remember to resolve all overfull or underfull warnings to help ensure your dissertation meets the margin requirements.[2] Click here to learn how to identify and resolve the warnings. Common causes of warnings include

- **LaTeX CANNOT automatically break long mathematical expressions or parenthetical references across multiple lines.** Chapter **??** demonstrates two methods for manually splitting equations.
- You can manually control the hyphenation by using the `\hyphenation{}` hints command. For example, write `\hyphenation{rep-re-sent-a-tive}` in the preamble to tell LaTeX where it can break the word "representative."
- The verbatim environment and the `\verb` command remove LaTeX's formatting controls. There may be instances that require you to manually adjust line breaks.
- The breaklinks option of the hyperref package is activated, however there may be instances of URLs running into the margins that require your manual intervention.

3.4 Template Tips

3.4.1 Chapters, sections, and subsections

Create chapters with the `\chapter` command, sections with the `\section` command, and subsections with the `\subsection` command. If you use the `\MakeUppercase` command to autocapitalize chapter titles, carefully check each title and its table of contents entry for compliance with University formatting requirements. You may wish to split chapters into separate .tex files for easier document management. You can use the `\include{}` command to add the chapters to your main .tex file. Click here to learn more.

If a chapter or section title contains math expressions or other symbols, the `hyperref` package may give a PDFDocEncoded warning when you compile your document. Use the `\texorpdfstring` command to edit the corresponding PDF bookmark text to resolve the warning. For example, the section title "Lemurs Wrote $E = mc^2$ in Mud on the Riverbank" should be coded

```
\section{Lemurs Wrote \texorpdfstring{$E=mc^2$}{E=mc2} in Mud on the Riverbank}
```

3.4.2 Page orientation

The template's default page orientation is portrait. Use the landscape environment to create landscape pages. Page number location and margins will automatically update like page **??**.

3.5 Getting Help and Giving Feedback

Questions about LaTeX?

- Click here to connect with LaTeX resources, tutorials, and more.

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[2] This is a common source of errors – please double check the margins before submitting your final dissertation.

Want to give feedback on the template?

- Click here to share it with us.

3.6 Acknowledgments

The University of Pennsylvania Ph.D. Dissertation LaTeX Template was initiated by Dr. Beth S. Wenger, Associate Dean for Graduate Studies, School of Arts and Sciences. It was developed by Lauren Gala, Penn Libraries with assistance from Dr. Julia Hartmann and Dr. Man Cheung Tsui, Department of Mathematics, and Bridget Rothera, Graduate Division School of Arts and Sciences. Portions of the code and text were reprinted or adapted with permission from Ratcliffe S.J. (2017) Penn Biostat LaTeX Template: PhD dissertation modified from Dissertation Template for Wharton PhD Candidates in LaTeX.

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TABLES AND FIGURES

4.1 Tables

The captions of tables inside the table environment (see Table **??**) will display in the List of Tables. Use the `\label` to label tables.

		Primes				
		2	3	5	7	
2*Powers	504	3	2	0	1	
	540	2	3	1	0	
2*Powers	gcd	2	2	0	0	min
	lcm	3	3	1	1	max

Tabel 4.1: Example Table

4.2 Figures

The captions of figures inside the figure environment (see Figure **??**) will display in the List of Illustrations. Use the `\label` to label figures.[1]

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[1] This also includes photographs, maps, schemes, diagrams, etc.

Gambar 4.1: Lemur photograph

LaTeX takes into account line and page breaks when determining where to place floating objects like figures and tables on a page. You can override LaTeX's placement decisions by using the float package options. Consult the float package documentation to learn how to gain more control over where and how float objects display.

Gambar 4.2: Lemur bar chart on landscape page

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EQUATIONS

The template automatically loads several math packages like amsmath, amssymb, and amsfonts. The split and the multline environments allow you to split an equation over multiple lines. Equation **??** is written in the split environment, and Equation **??** is in the multline environment. Notice the differences in how the two environments align the equations and where they place the equation number.

$$A = \frac{\pi r^2}{2} = \frac{1}{2}\pi r^2 \tag{5.1}$$

$$\begin{multline*}\mathrm{p(x)} = 3\mathrm{x}^6 + 14x^5y + 590x^4y^2 + 19x^3y^3 + 3x^6 + 14x^5y + 590x^4y^2 + 19x^3y^3 \\ + 3x^6 + 14x^5y + 590x^4y^2 + 19x^3y^3 - 12x^2y^4 - 12xy^5 + 2y^6 - a^3b^3\end{multline*}$$

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APPENDICES

Appendices are optional. Use the `\chapter` command inside the custom `append` environment to insert appendices. Use the `\ref{ }` command to reference appendices in your text.

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CITATIONS AND BIBLIOGRAPHY

LaTeX bibliography packages and citation management programs enable you to efficiently cite and format bibliographic references. You can manage bibliographic entries in a separate bibliography database file (*.bib) or you can embed the entries directly into the main .tex file. While the LaTeX template was built for the former approach, you can modify it to accommodate latter.

7.1 Getting Started

Please watch this short video to learn the principles underpinning the template bibliography before proceeding.

7.2 Default Template Settings

The commands written in sampledissertation.tex load the natbib package in its preamble and reference the bibliographic entries stored in bibliography.bib.

This is an example inline citation heard$_{c}harles_{2}020.Ifyouwanttocitetwopublicationsinoneplacelikemcgoogan_{c}hasing$

7.3 Customization

Citation and bibliography formatting commands are written in the main .tex file for your convenience. You can edit the relevant code in the .tex file to produce citation and bibliography styles appropriate for your field. If you wish to make significant to changes to the structure of the bibliography, you may need to edit the bibliof environment defined in upennstyle.sty.

7.3.1 Styles and formatting

The natbib package, selected format and style options, and example citations are for demonstration. In addition to natbib, other packages and accompanying documentation are free to download from CTAN. Always consult the package documentation for additional citation and bibliography commands and formatting options. Remember that the available commands and styles differ between packages.

7.3.2 Embedded bibliography

You can edit the template to manage your bibliography directly in the .tex file obviating the need for a separate .bib file. Replace the `\bibliography` command in the bibliof environment with the list of bibitems and edit relevant code elsewhere in the template as needed.

7.4 Recommended Resources

- Click here to learn how to manage citations and bibliographies in LaTeX documents
- Click here to learn citation best practices and how to use citation management programs
- Click here to watch a short introduction to citing and formatting references with natbib

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TITLE OF APPENDIX A

The content of Appendix A begins here. Use the `\chapter` command to insert additional appendices.

8.1 Section Name of Appendix A

Use the `\section` command to create sections.

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TITLE OF APPENDIX B

The content of Appendix B begins here.